

Arithmetic operations

- $\left. \begin{array}{l} \rightarrow + \text{ addition} \\ \rightarrow - \text{ subtraction} \\ \rightarrow * \text{ multiplication} \\ \rightarrow / \text{ division} \end{array} \right\}$

for blocks of a few bits
Byte level $\rightarrow$ 8 bits at a time

Steps

① byte $\rightarrow$ represent it as a polynomial

② choose a prime polynomial of degree "8"

$$\underline{a(x)b(x) \bmod p(x)} \rightarrow \text{9 deg at most}$$

$$1/a(x) = ?$$

$$a(x) \underline{\underline{\bar{a}(x)}} \bmod p(x) = 1$$

Example

$$a = 00010001$$

$$b = 00011001$$

1) $a+b$, 2) $a-b$, 3) $a*b$, 4) a/b

Addition

$$a \rightarrow a(x) = x^4 + 1$$

$$b \rightarrow b(x) = x^4 + x^3 + 1$$

$$a(x) + b(x) = \cancel{x^4} + \cancel{x^4} + \cancel{x^3} + \cancel{x^3} + 1 + 1$$

$$= x^3$$

$$a+b = 00001000$$

2) $a-b$ $a(x) = x^4 + 1$
 $b(x) = x^4 + x^3 + 1$

$$\underline{a(x) - b(x)}$$

$$\cancel{x^4} + \cancel{x^4} + \cancel{x^3} + \cancel{x^3} + 1 - 1$$

$$= x^3$$

3) $a*b$

$$= a(x) * b(x) \bmod p(x) \text{ --- ①}$$

$$p(x) = x^8 + x^4 + x^3 + x + 1$$

$$(x^4 + 1) * (x^4 + x^3 + 1) \bmod p(x)$$

$$\cancel{x^8} + \cancel{x^7} + \cancel{x^4} + \cancel{x^4} + x^3 + 1$$

$$(x^8 + x^7 + x^3 + 1) \bmod \underline{\underline{x^8 + x^4 + x^3 + x + 1}}$$

$$x^8 + x^4 + x^3 + x + 1$$

$$\begin{array}{r} \cancel{x^8} + \cancel{x^7} + \cancel{x^3} + \cancel{x} + \cancel{1} \\ \underline{} \\ x^4 + x^7 + x^3 + x \end{array}$$

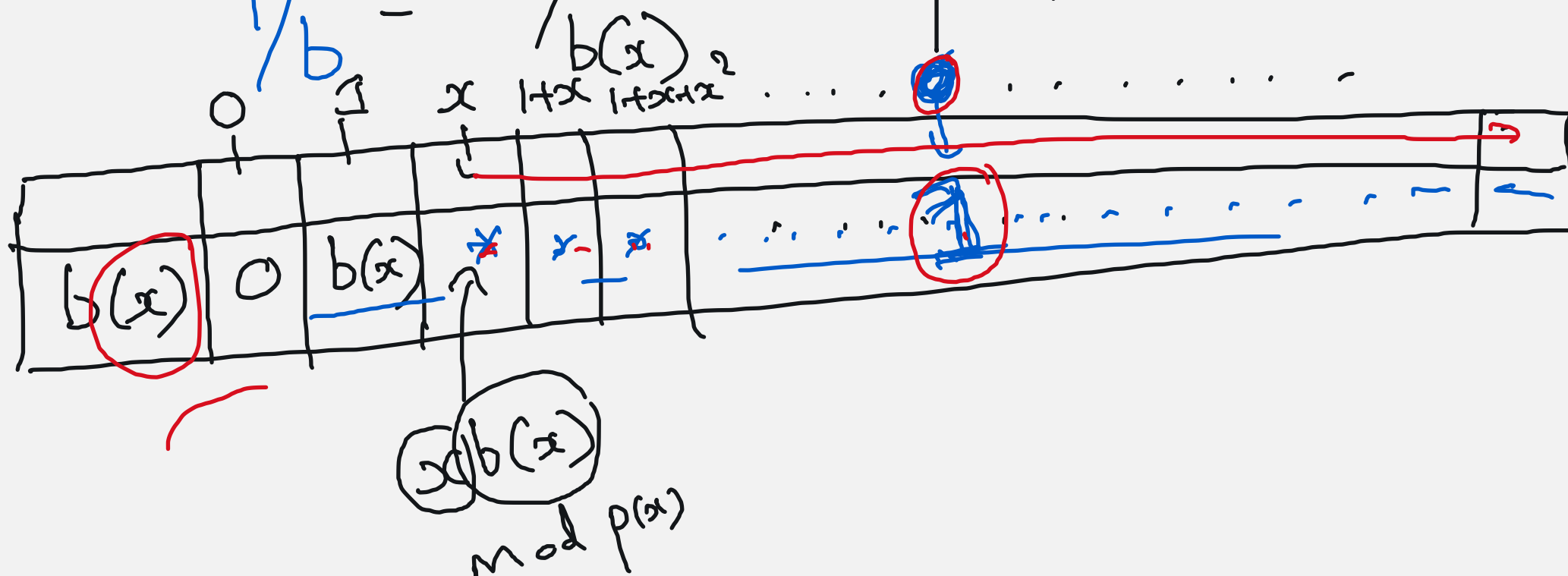
$$x^4 + x^7 + x^3 + x$$

$$\boxed{10010010}$$

$$a/b = a * (1/b)$$

inverse of b

$$1/b = 1/b(x) \bmod p(x)$$



$$\underline{\underline{x b(x) \bmod p(x)}}$$

$$\underline{a(x) * (1/b(x))}$$

$$1/b(x) ? \quad (b(x) * ?) \bmod p(x) = 1$$